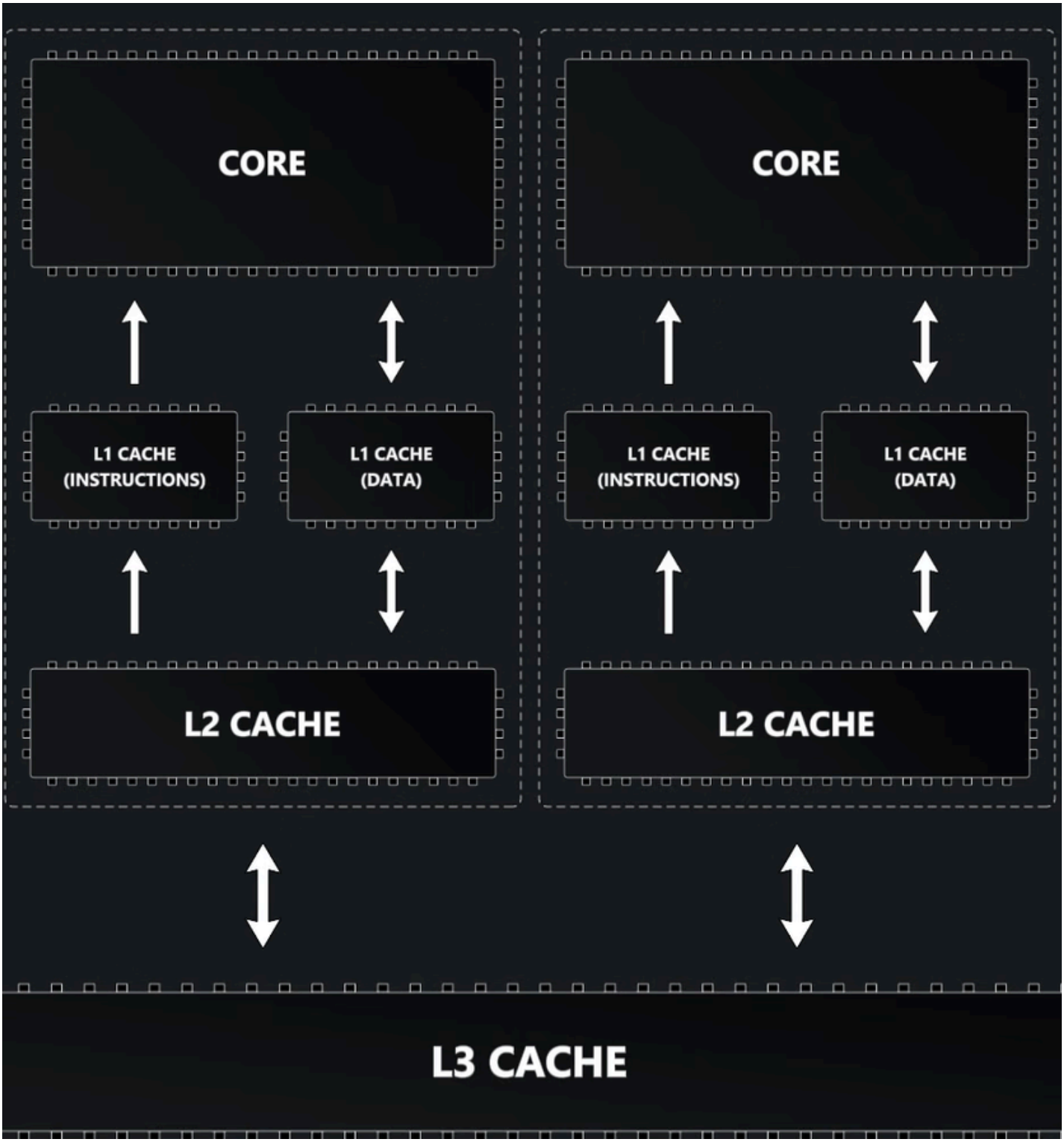
Rana Ahmed Elmaghraby

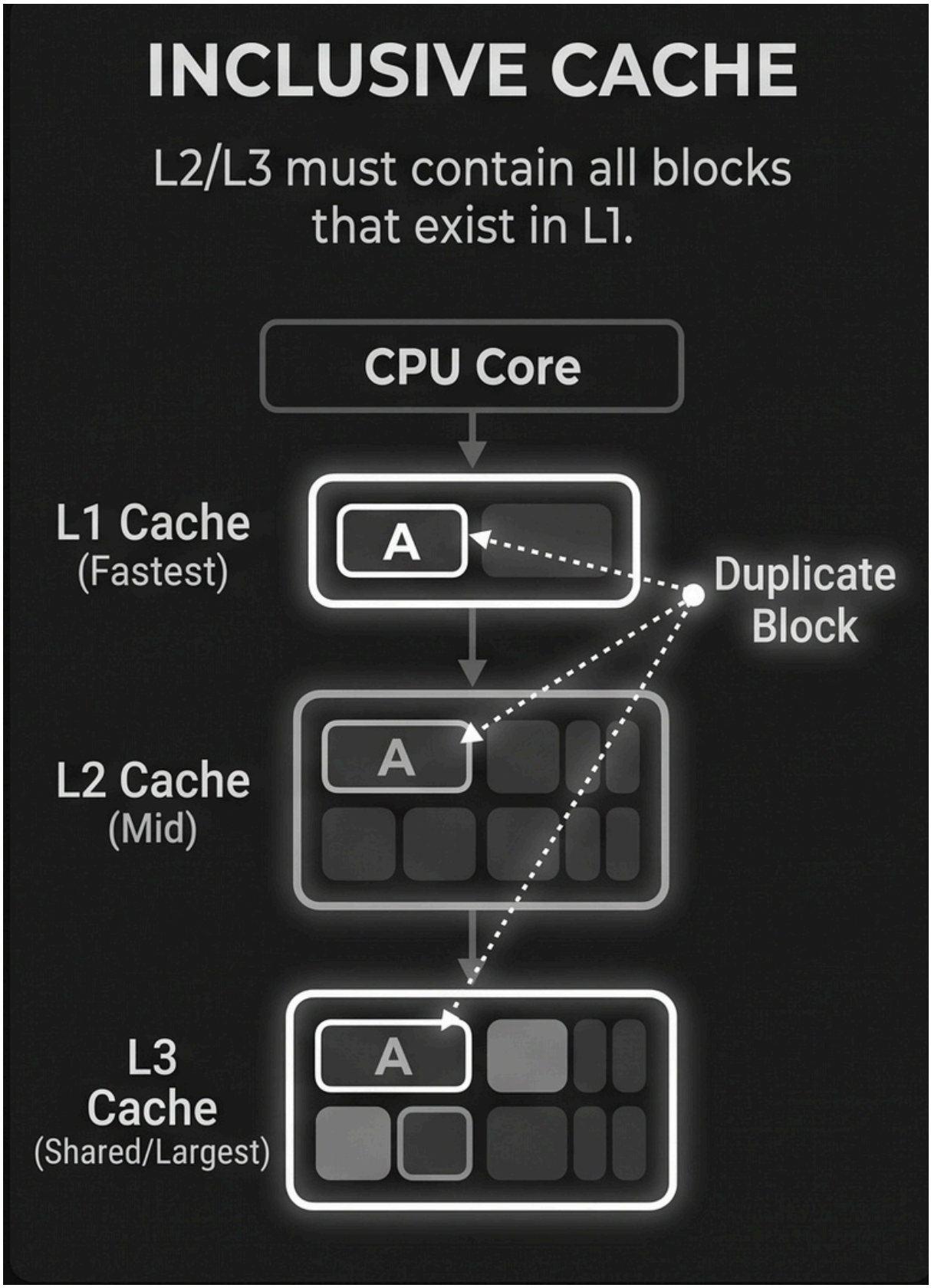
Task 1 (HPC)
Level: Third Year
Department: AI (General)

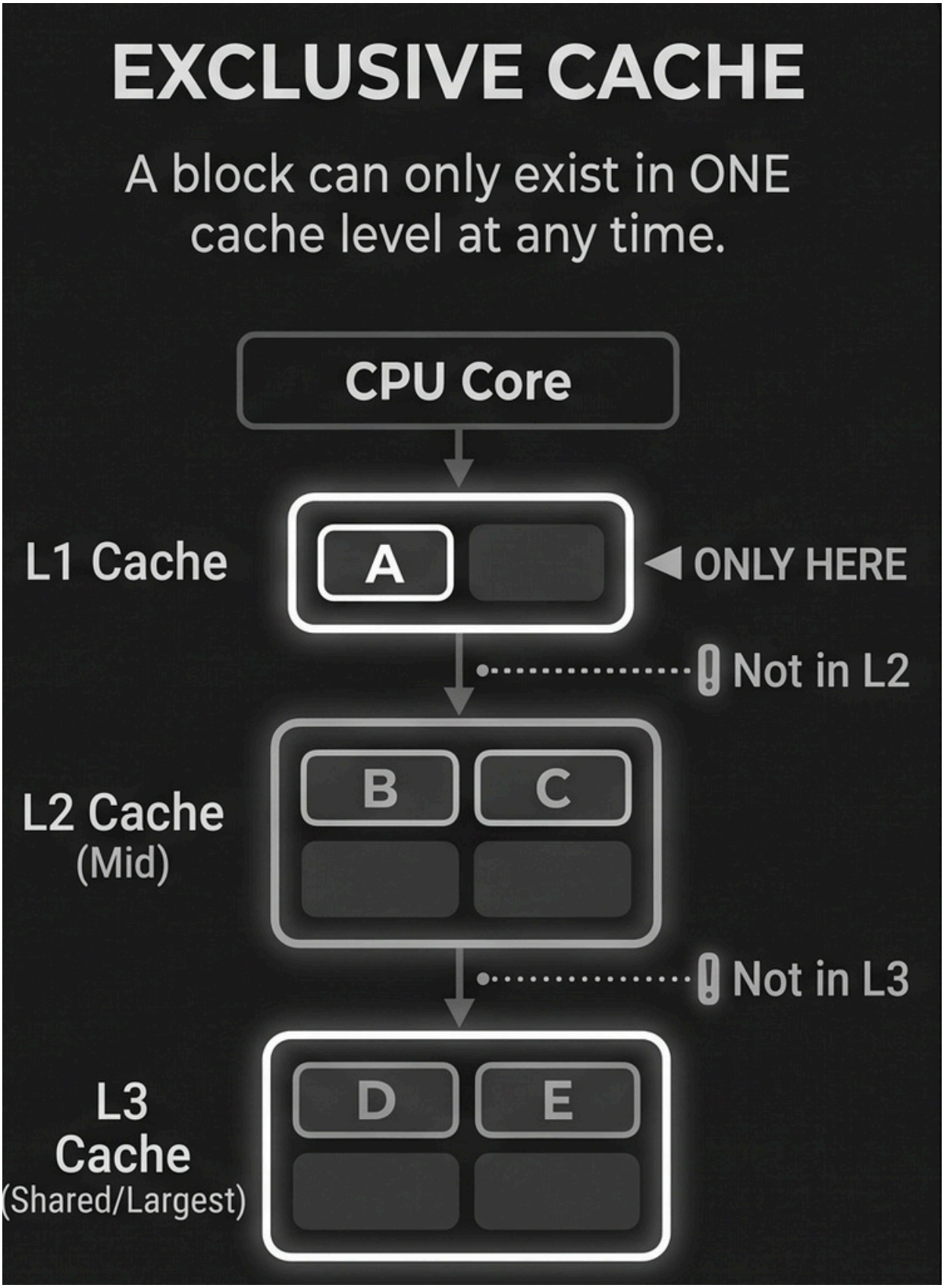
CPU Cache Hierarchy

Cache Level	Typical Size	Latency (Cycles)	Associativity	Key Characteristics
L1 Cache	16 KB – 128 KB (per core)	~1 – 4 cycles	2-way to 8-way	Fastest & smallest. Divided into two separate components (Split Cache): L1d (for Data) and L1i (for Instructions).
L2 Cache	256 KB – 2 MB (per core)	~4 – 10 cycles	4-way to 16-way	Dedicated to a single core. Acts as a Unified Cache (stores both data and instructions) and communicates directly with L1.
L3 Cache	2 MB – 32+ MB (shared)	~10 – 40 cycles	~16-way	Largest & slowest cache. Shared across all CPU cores to allow fast inter-core data sharing without accessing RAM.


CORE
CORE
L1 CACHE (INSTRUCTIONS)
L1 CACHE (DATA)
L1 CACHE (INSTRUCTIONS)
L1 CACHE (DATA)
L2 CACHE
L2 CACHE
L3 CACHE

Policy / Step	Description
Inclusive Cache	Every block in a higher-level cache must also exist in the specified lower-level cache. For example, a block in L1 must also exist in L2 or L3, depending on the design.
Exclusive Cache	A cache block exists in only one of the relevant cache levels at a time.
Non-Inclusive, Non-Exclusive (NINE)	There is no strict rule requiring duplication or preventing it; a block may exist in multiple levels or only one.


INCLUSIVE CACHE
L2/L3 must contain all blocks that exist in L1.
CPU Core
L1 Cache
(Fastest)
A
Duplicate Block
L2 Cache
(Mid)
A
L3 Cache
(Shared/Largest)
A

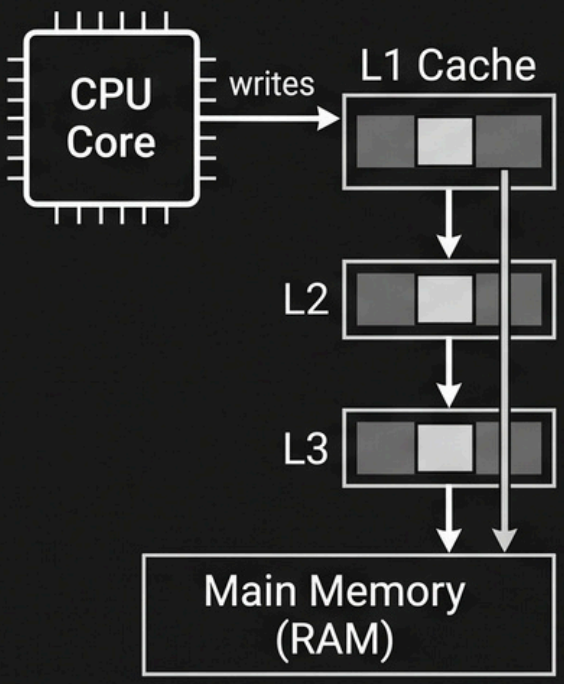

EXCLUSIVE CACHE
A block can only exist in ONE cache level at any time.
CPU Core
L1 Cache
A
ONLY HERE
Not in L2
L2 Cache
(Mid)
B
C
Not in L3
L3 Cache
(Shared/Largest)
D
E

Category / Policy	Type / Name	Description & Implementation
Write Operation (Write Hit)	Write-Through	Every write updates the cache and propagates the update to the next memory level. Simplifies keeping lower-level copies updated but may increase memory traffic.
Write Operation (Write Hit)	Write-Back	The modified cache block is marked dirty. The update is written to a lower level later (typically when evicted). Reduces immediate write traffic but requires tracking modified blocks.
Write Miss Policy	Write-Allocate	The cache block is brought into the cache, then modified. It is commonly paired with write-back.
Write Miss Policy	No-Write-Allocate	The block is not brought into that cache; the write is directed to a lower memory level. It is commonly paired with write-through.

WRITE HIT POLICIES
(Data is in Cache)

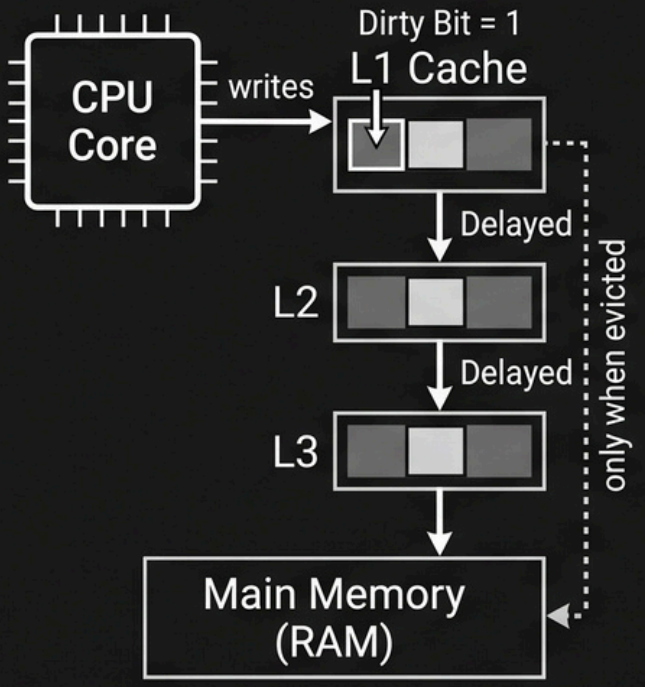
How updates are made when the target address is found in the cache (e.g., L1).

1A. WRITE-THROUGH



- Updates the cache and lower memory levels immediately.
- Ensures complete consistency.
- Higher latency due to slow RAM writes.

1B. WRITE-BACK

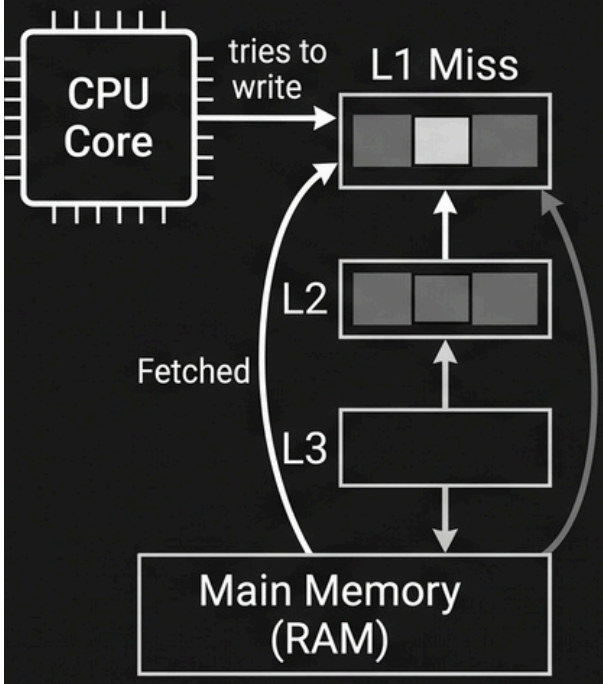


- Updates L1 cache only (marks block as 'Dirty').
- Lower levels & RAM are updated later (only when the block is
- Lower latency, reduces memory bus traffic.

WRITE MISS POLICIES
(Data is NOT in Cache)

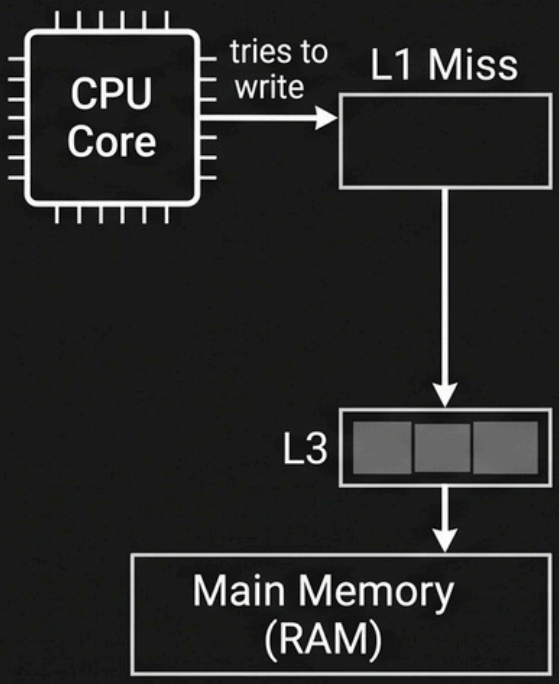
How the system responds when the target address is not found in the cache.

2A. WRITE-ALLOCATE
(Fetch-on-Write)



- Brings the missing block into the cache first, then modifies it.
- Typically used with Write-Back.

2B. NO-WRITE-ALLOCATE
(Write-Around)



- Bypasses the cache; writes directly to a lower level or RAM.
- Block is not loaded into the cache.
- Typically used with Write-Through.